Computational Methods of Quantile Regression

Roger Koenker
University of Illinois at Urbana-Champaign

CEMMAP/UCL: February, 2003

The Gaussian Hare vs the Laplacian Tortoise

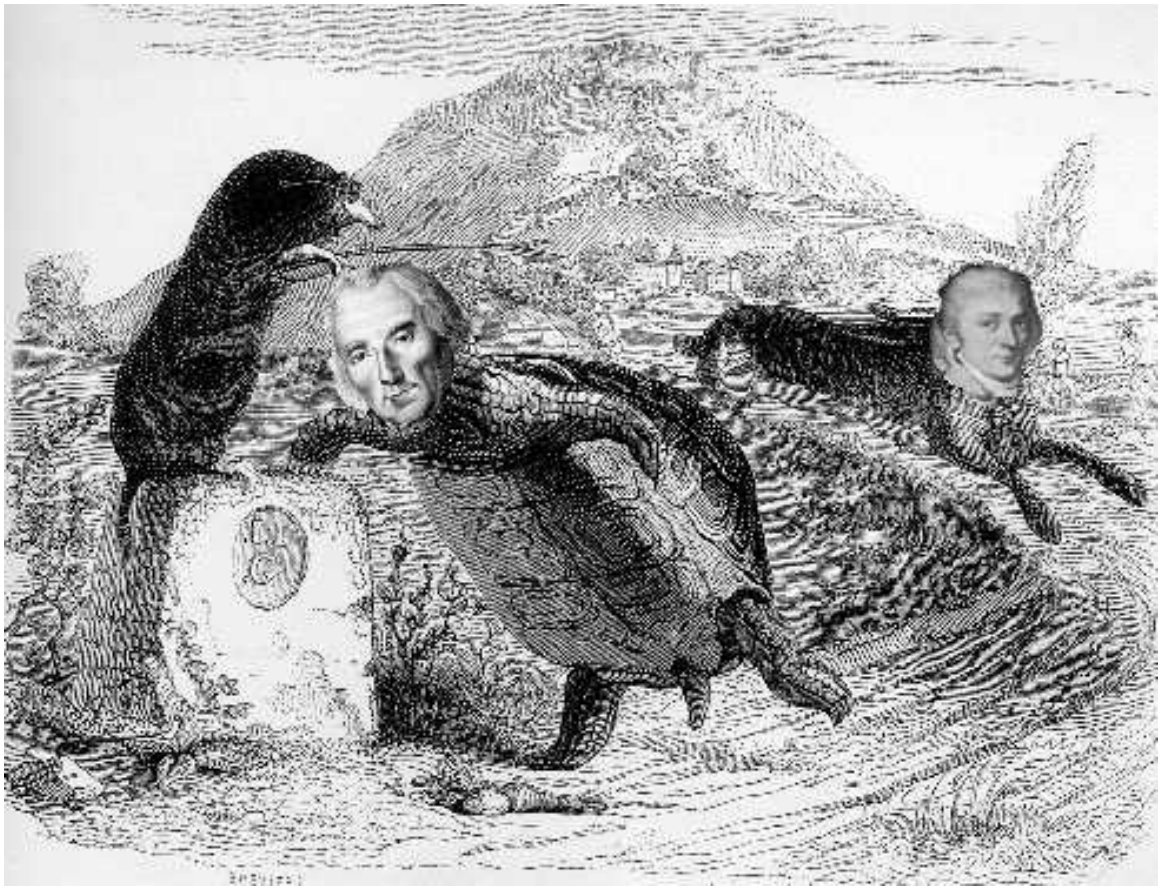


Figure 1: Retouched 18th century woodblock photo-print

Outline

- Boscovich's Travails
- Laplace's Methode de Situation
- Edgeworth's Dual Plot and Plural Median
- Dantzig's Simplex Method and the Cauchy Algorithm
- Frisch's Interior Point Radar

Boscovich's Problem

Boscovich (1757) proposed estimating the ellipticity of the earth by solving:

$$\min \sum_{i=1}^{n} |y_i - \alpha - x_i'\beta|$$

subject to:

$$\sum_{i=1}^{n} (y_i - \alpha - x_i'\beta) = 0,$$

The constraint allowed the problem to be reduced to the weighted median problem:

$$\min \sum_{i=1}^{n} |\tilde{y}_i - \tilde{x}_i'\beta|$$

where $\tilde{y}$ and $\tilde{x}$ denote deviations from their means.

Laplace's *Methode de Situation*

Laplace showed that Boscovich's problem:

$$\min \sum_{i=1}^{n} |\tilde{y}_i - \tilde{x}_i' \beta|$$

could be solved by ordering the candidate slopes $\{\tilde{y}_i / \tilde{x}_i\}$ and finding the weighted median using weights $w_i = |\tilde{x}_i|$, i.e. finding the smallest j such that,

$$\sum_{i=1}^{j} w_{(i)} > \frac{1}{2} \sum_{i=1}^{n} w_{(i)}.$$

An Historical Prelude: The Ellipticity of the Earth

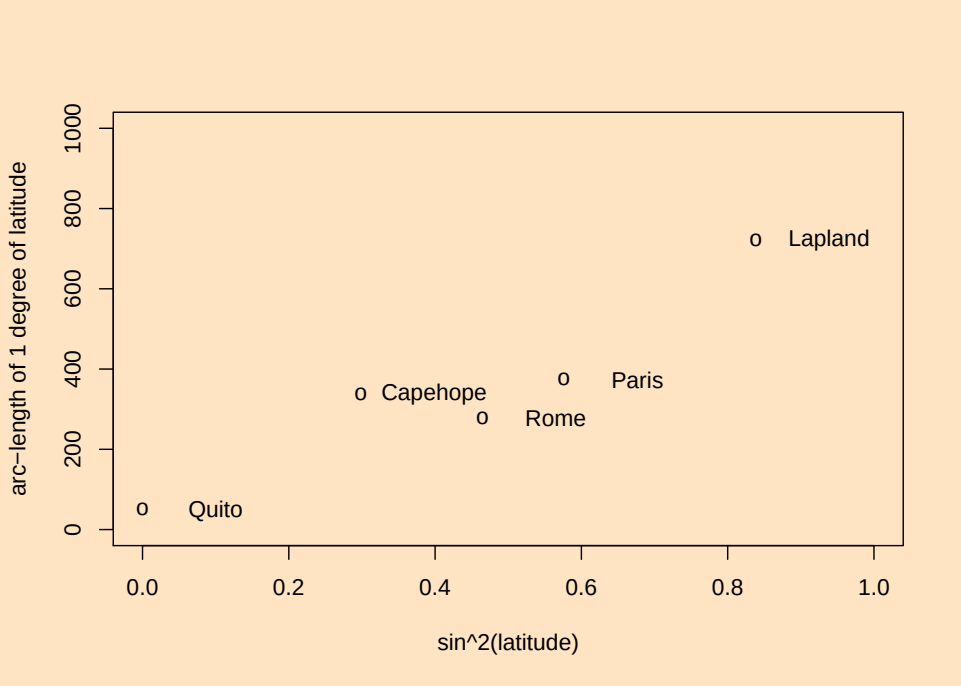


Figure 2: Grapefruit or Lemon? Boscovich (1755) considered the five measurements depicted above. Arc length is measured in toise per degree of latitude minus 56,700. An upward slope in this figure indicates that the earth is oblate (like a grapefruit) rather than prolate (like a lemon).

The First Regression

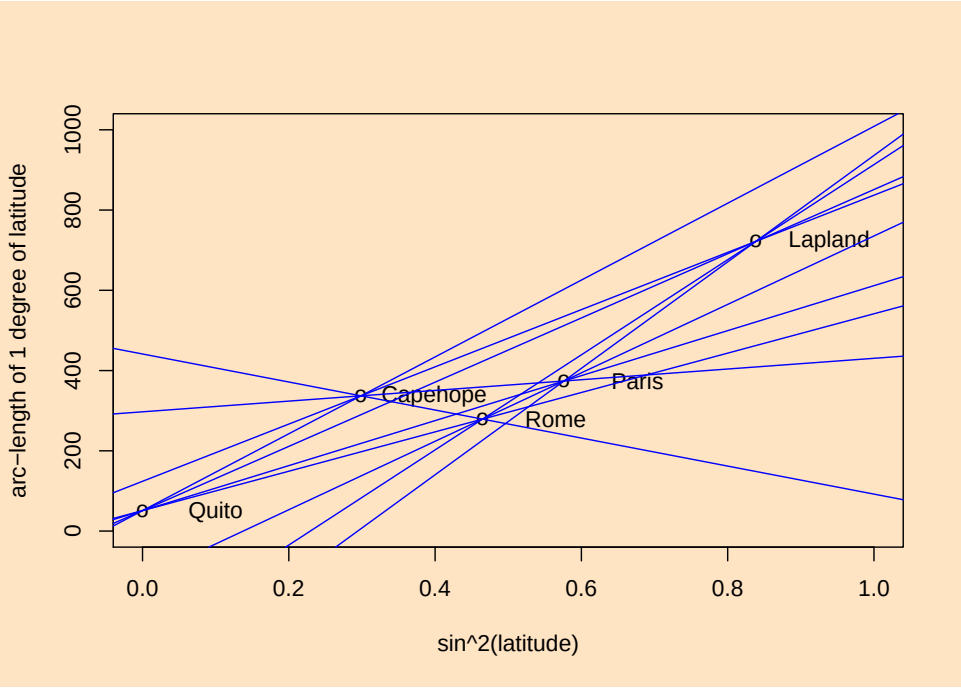


Figure 3: Boscovich (1755) computed all the pairwise slopes and initially reported a trimmed mean of the pairwise slopes as a point estimate of the earth's ellipticity.

The Second Regression

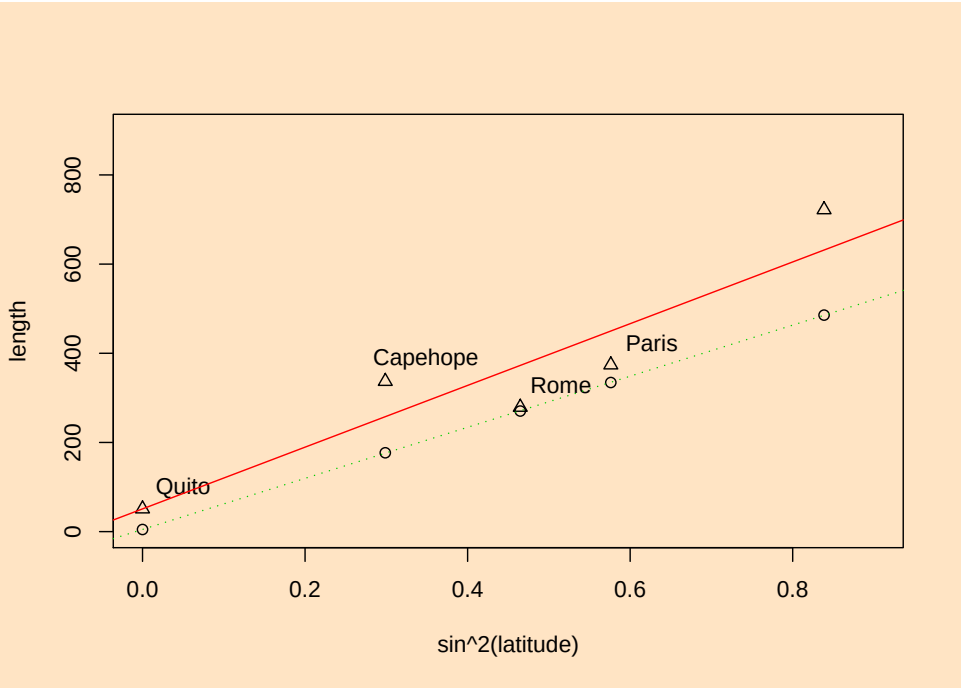


Figure 4: In 1757 Boscovich reestimated the model minimizing the sum of absolute residuals subject to the constraint that the residuals summed to zero to obtain the solid line fitting the triangular points. Modern measurements of the arc lengths are represented by the circles.

Earth's Ellipticity: A Quantile Regression View

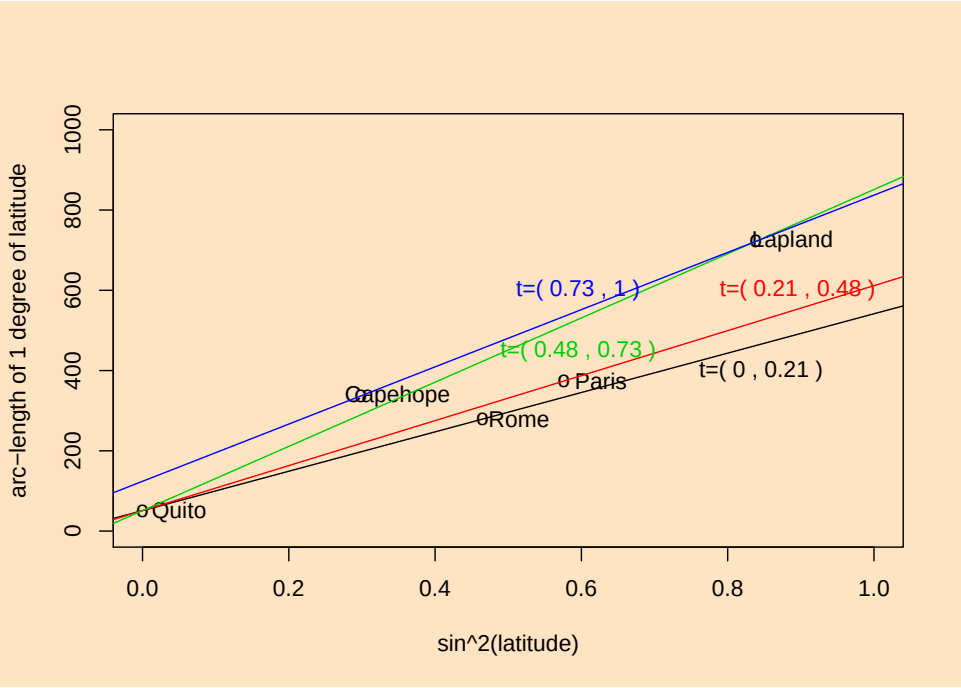


Figure 5: The quantile regression analysis of the Boscovich data identifies four distinct pairs of points that solve the weighted ℓ_1 problem for various intervals of the parameter τ.

Edgeworth's Dual Plot and the Plural Median

F.Y. Edgeworth (1888) suggested solving

$$\min \sum_{i=1}^{n} |y_i - \alpha - x_i'\beta|$$

without Boscovich's constraint. His computational strategy introduced the "dual plot". Each pair of observations defined a point in parameter space, and lines in this space represent exact fits to one of the sample observations. Edgeworth suggested that a solution could be found by following a path of steapest descent through the dual plot. The multivariate version of this required "some power of hypergeometrical conception."

Dantzig's Simplex Method and the Cauchy Algorithm

Dantzig's (1948) simplex method for linear programming provided a general method for Edgeworth's plural median problem. points in the general p dimensional setting corresponded to basic solutions, exact fits of p observations, and lines corresponded to solutions with $p-1$ zero residuals. Traversing the dual plot by steapest descent is the simplex method. Barrodale and Roberts (1978) algorithm implemented a particularly efficient version that can be interpreted as a form of the Cauchy algorithm: steapest descent at each step with one dimensional line search in the descent direction.

Frisch's Interior Point Radar

In the early 1950's Ragnar Frisch suggest that logarithmic barrier methods could be used for linear programming problems:

> My method is altogether different than simplex. In this method we work systematically from the interior of the admissible region and employ a logarithmic potential as a guide – a sort of radar – in order to avoid crossing the boundary.

This method anticipates modern "interior point" methods for linear programming.

Interior versus Exterior Point Methods

Problem: Find the most northeast point in a circumscribed polygon. Formally:

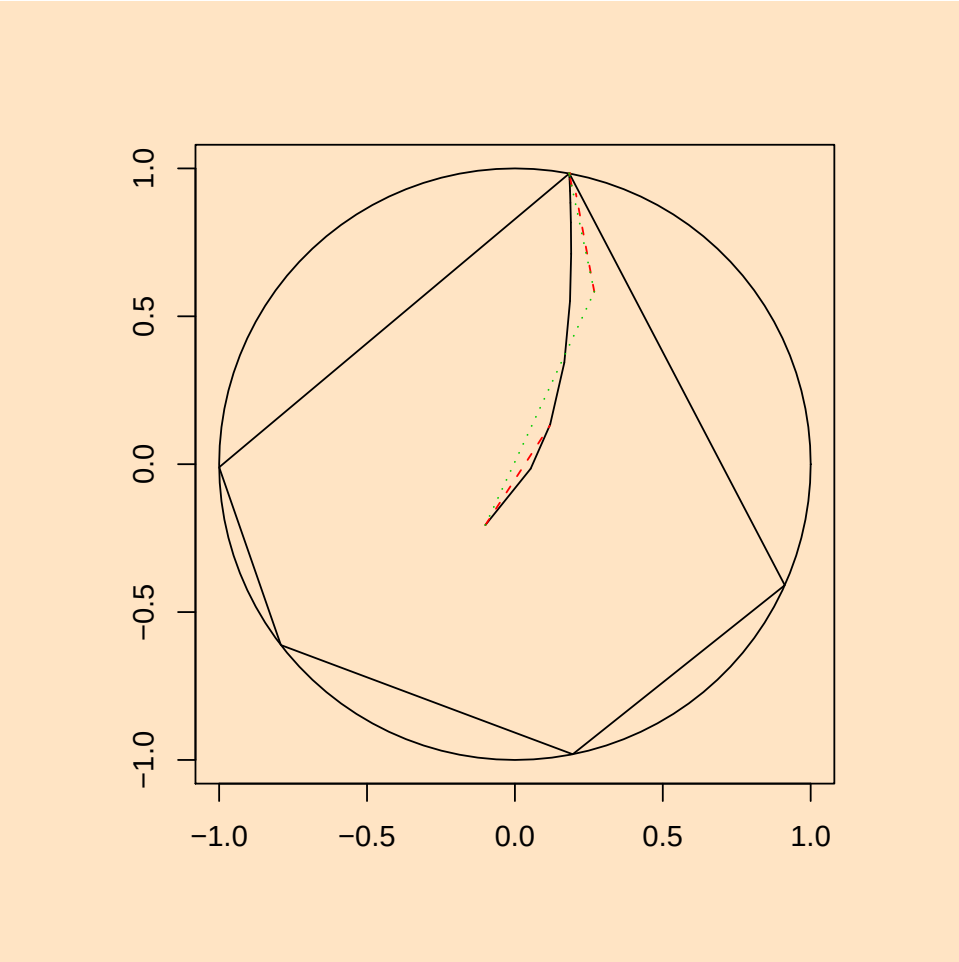
$$\max\{e'u | A'd = u,\ e'd = 1,\ d \geq 0\}$$

were e is vector of ones, and A has rows representing the n vertices. Eliminating u we can reformulate the problem as:

$$\max\{c'x | e'x = 1,\ \ x \geq 0\},$$

where $c = Ae$. Simplex is very simple here: we start at a random vertex and go around the edge of polygon until we find the optimum.

A Simple LP Problem



Interior versus Exterior Point Methods

In the interior point method we replace

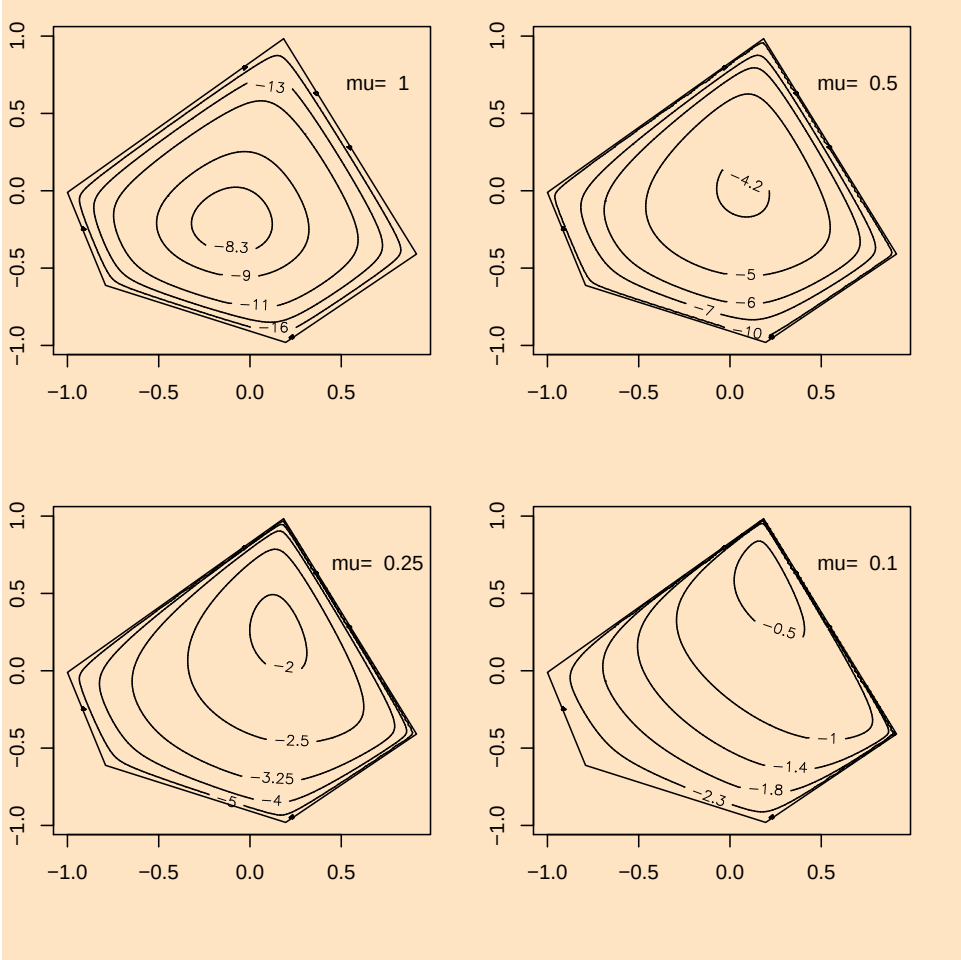
$$\max\{c'x | e'x = 1, \;\; x \geq 0\},$$

with the log barrier reformulation

$$\max\{c'x + \mu \sum_{i=1}^{n} \log x_i | e'x = 1\}$$

Judiciously, letting $\mu \to 0$ leads us to a solution. The curved central path in the prior figure illustrates the locus of solutions as μ tends to zero.

Contours of the Log Barrier Objective Function



Interior Point Formulation of Quantile Regression

The dual formulation of the quantile regression problem is,

$$\max_a \{y'a \mid X'a = (1-\tau)X'e, \;\; a \in [0,1]^n\},$$

Again we simply replace the inequality constraints with the barrier penalty

$$\max_a \{y'a + \mu \sum (\log a_i + \log(1-a_i)) \mid X'a = (1-\tau)X'1\},$$

and proceed. In the new formulation we can compute Newton steps; Portnoy and Koenker (1997) call this a Frisch-Newton algorithm. With some strategic preprocessing timings are very competitive with least squares even on extremely large problems.

Postscript on Computation

Returning to Edgeworth's dual plot we may remark that the deepest point in the dual plot in the sense of Rousseeuw and Hubert (1999). This leads to a very interesting high-breakdown form of quantile regression as they discuss:

$$\sup_{\beta \in \mathbb{R}^p} \inf_{||\gamma||=1} \sum \psi_\tau(y_i - x_i^\top \beta) \mathrm{sgn}(x_i^\top \gamma),$$

This is extremely challenging from a computational viewpoint, and very intriguing from a statistical viewpoint.

Laplacian Tortoise versus Gaussian Hare

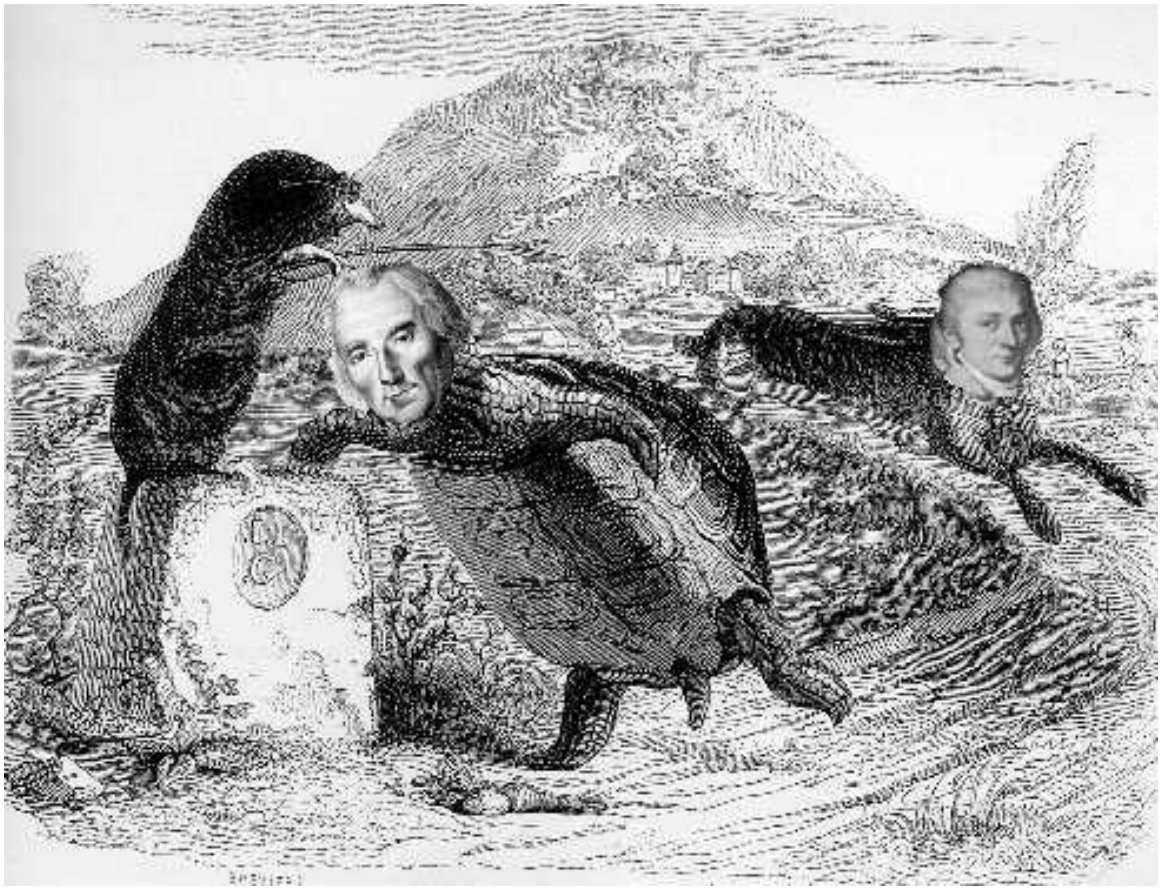
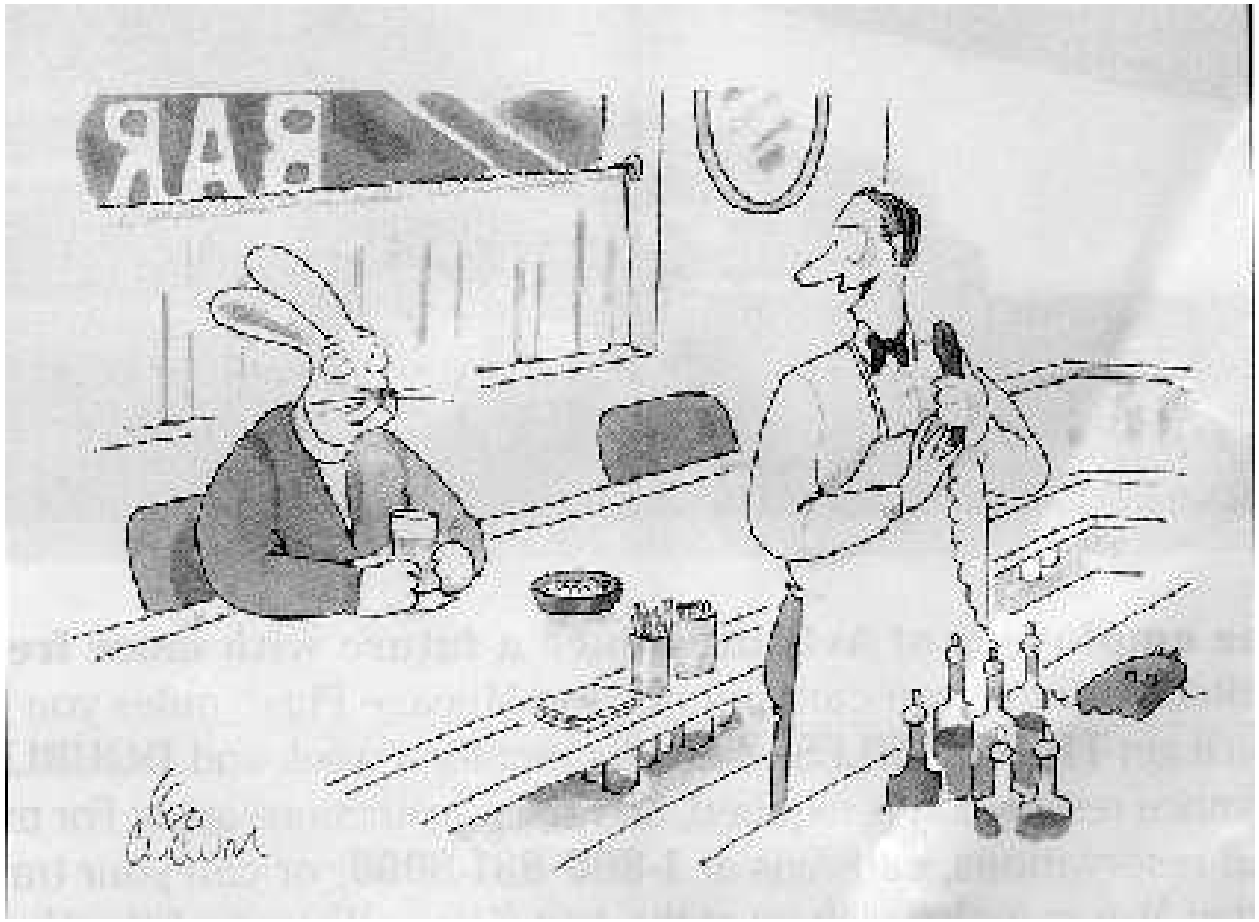


Figure 6: Retouched 18th century woodblock photo-print

Another Interpretation



"It's about the tortoise ... he's tested positive for steroids."